STOCKSENSE HARDWARE & SYSTEM ARCHITECTURE

Component Shortlist & Technical Justifications

Document ID	Author	Target PCB
SPEC-HW-BOM	**Kavindu Hasaranga**	**Rev A (2-Layer SMD)**

1. Microcontroller (MCU) Core

Specification	Primary Choice: ESP32-WROOM-32U	Candidate Option: STM32F401 / STM32L4
Core Architecture	32-bit Dual-Core Xtensa LX6 @ 240 MHz	32-bit ARM Cortex-M4 @ 84 MHz
Memory	520 KB SRAM, 4 MB / 8 MB SPI Flash	96 KB SRAM, 512 KB Flash
Wireless	Integrated Wi-Fi (802.11 b/g/n) & BLE v4.2	External Module (ESP8266 via AT Commands)

Technical Justification:

- **TLS Hardware Acceleration:** Native mbedTLS stack offloads RSA/AES cryptographic handshakes required for MQTTS (port 8883) without disrupting real-time ADC sampling.
- **Dual-Core Task Separation:** FreeRTOS pins network stack and MQTT keep-alives to Core 0 while Core 1 handles strain gauge filtering and interrupt handlers.
- **External U.FL Antenna (WROOM-32U):** Industrial metallic rack environments attenuate PCB trace antenna signals; external rubber-duck antennas ensure reliable connectivity.

2. Analog Front-End (AFE) & Load Cell Sensor

Specification	Primary Choice: HX711 ADC	Secondary Choice: ADS1232
Resolution & PGA	24-bit Sigma-Delta, Selectable Gain (64/128)	24-bit Low-Noise, Programmable Gain (1–128)
Sample Rate	10 Hz or 80 Hz (pin selectable)	10 Hz or 80 Hz
Sensor Element	Single-Point Parallel Beam Strain Gauge (5 kg / 20 kg / 50 kg Wheatstone Bridge)	

Technical Justification:

- **Sub-Gram Accuracy:** 24-bit dynamic range provides >16 million steps, allowing system to easily detect single-item weight deltas (±1% accuracy goal).
- **Differential Input Channels:** Channel A matches Wheatstone bridge sensitivity (2 mV/V) directly without additional op-amps.

3. Identification & Environmental Sensors

Sensor Role	Selected Component	Bus Interface & Key Parameters
RFID Reader	**MFRC522 Module**	13.56 MHz (SPI bus up to 10 Mbit/s), ISO/IEC 14443A tag support (<50 ms scan).
Climate Monitor	**SHT30 / SHT31**	I²C interface (address 0x44), ±2% RH, ±0.2°C accuracy for perishable inventory.

4. Power Management Subsystem (PMIC & Battery Backup)

Subsystem	Selected Component	Role & Operating Characteristics
Input Power	12V DC Barrel Jack / Terminal Block	Regulated 12V DC primary supply rail.
Charger & Power Path	**TP5100 / CN3791 + Diode-OR**	Buck Li-Ion charger configured for 3S (12.6V max) with dual-diode PowerPath.
Backup Battery	3S 18650 Pack (11.1V nominal, 2600 mAh)	Sustains full node power during mains power outage (≥8 hours).
Battery Protection	**3S 18650 10A–20A BMS**	Integrated 3S BMS for over-charge, over-discharge, short-circuit protection, and cell balancing.
Buck Regulator	**MP2307 / LM2596-5.0**	Steps down 12V rail to 5V DC for high-current peripherals.
3.3V System LDO	**AP2112K-3.3 / AMS1117-3.3**	Steps down 5V to clean 3.3V DC for ESP32 and digital sensors.

Technical Justification: The TP5100/CN3791 provides multi-cell buck charging for the 3S Li-ion battery from the 12V supply. The dual-diode PowerPath ensures that when external 12V power is disconnected, power automatically routes from the 3S battery to the MP2307 buck converter without MCU reset or telemetry drop.

5. Prototyping Gear & Interconnects (Bench Phase)

Item Category	Specific Item Description	Application / Quantity per Bench Setup
Breadboards	830-Point Solderless Breadboards with Power Rails	2 units for sensor & MCU prototyping workspace.
Jumper Wires	28 AWG Flexible Ribbon Jumpers (M-M, M-F, F-F 20 cm)	1 box (120 wires total) for pin-to-pin breadboard setup.
Pin Headers	2.54 mm Pitch Breakaway Male & Female Headers	20 strips (40-pin) for breakout board socketing.
Terminal Blocks	5.08 mm Screw Terminal Blocks (2-Pin & 4-Pin)	4 units for secure load cell wire (E+, E-, A+, A-) connection.
Power Interconnect	USB-C Female Breakout Board (5V/GND Pins)	2 units for powering bench breadboards cleanly.

6. Discrete Passive Components & Semiconductors (PCB Phase)

Component Type	Value / Part Number	Package / Footprint	Circuit Function & Placement
Resistors	10 kΩ ±1%	0805 / 1/4W PTH	I²C bus pull-ups (SDA/SCL) and ESP32 EN pull-up.
	100 Ω ±1%	0805 / 1/4W PTH	Low-pass RC filter on HX711 differential inputs (INA+/INA-).
	220 Ω / 1 kΩ	0805 / 1/4W PTH	Current limiting for status LEDs and MOSFET gate drive.
	30 kΩ & 10 kΩ	0805 / 1/4W PTH	Precision divider for battery & 5V mains sense GPIO pins.

6. Discrete Passive Components & Semiconductors (Continued)

Component Type	Value / Part Number	Package / Footprint	Circuit Function & Placement
Capacitors	100 nF (0.1 µF) Ceramic	0805 / Ceramic MLCC	Decoupling capacitors placed close to VCC pins of all ICs.
	10 µF to 100 µF 16V	Electrolytic / 1210	Bulk decoupling on 5V and 3.3V power rails for ESP32 Wi-Fi bursts.
	10 nF Ceramic	0805 MLCC	Differential input filtering capacitor across AFE input channels.
Diodes & Transistors	SS14 / 1N5819 Schottky	SMA / DO-41	Reverse polarity power supply input protection.
	TVS Diode Array	SOT-23	ESD protection on USB-C input and external sensor lines.
	AO3400A / 2N7002 MOSFET	SOT-23 / TO-92	Low-side N-Channel switches for status buzzer/relay control.
Mechanical	M3 Brass Standoffs (12mm)	M3 Screw Mount	PCB clearance mounting inside shelf-mounted enclosure.

7. Cloud Infrastructure & Software Stack

Layer	Selected Technology	Technical Rationale & Role
MQTT Broker	**EMQX / Mosquitto**	High-throughput broker supporting TLS 1.2/1.3, dynamic ACLs, and shared subscriptions.
Time-Series DB	**TimescaleDB**	PostgreSQL extension optimizing continuous telemetry ingests while preserving relational tables.
Backend Framework	**FastAPI (Python)**	Asynchronous REST and WebSocket engine delivering live telemetry feeds to user web app.
Web Frontend	**React.js + Tailwind**	Responsive component dashboard supporting trilingual localization (English, Sinhala, Tamil).
Analytics & Forecasting	**Python Worker**	Runs Holt-Winters time-series algorithms for consumption forecasting and reorder threshold alerts.
Deployment	**Docker & VPS**	Containerized deployment ensuring identical development and production environments.

Summary of BOM Completeness: The combination of core active ICs, passive discrete components (resistors, MLCC capacitors), interconnects, and protection devices ensures full compliance with both prototyping bench testing and custom PCB fabrication requirements.